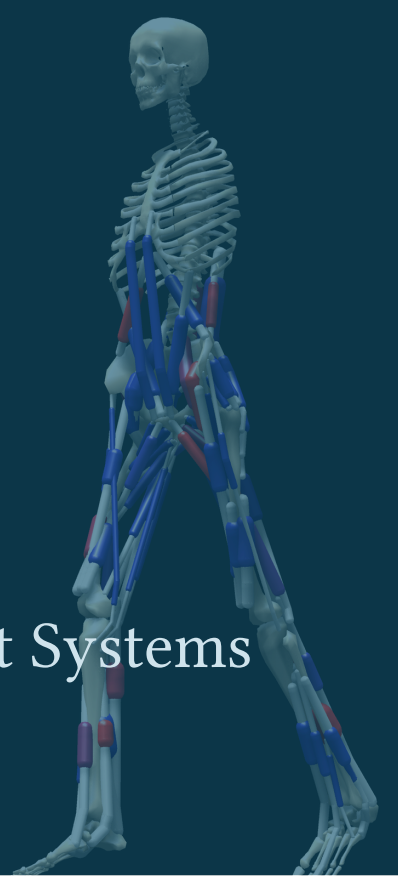


Stochastic Decision Horizons for Survival-Based Constrained Reinforcement Learning

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Constraint-dependent continuation is common in RL and often looks like a heuristic. We identify the exact survival-chance problem it solves, derive replay-compatible off-policy algorithms, and show when its statistic does, or does not, track cumulative cost.

1 - Motivation

- **Budgets:** fuel / money $\rightarrow$ cumulative-cost CMDPs.
- **Local operating limits:** deteriorating behavior, not spending transferable resources.
- **Continuation (e.g., CaT):** per-step violations shorten the effective decision horizon.
- **What constrained objective? Why can dual optimization sometimes be avoided?**

2 - The problem behind survival-based constraints

Fold all violations into a continuation probability $\alpha_\lambda(s, a) = \exp\{-\lambda(c_1 + \dots + c_n)\}$, driving Bernoulli continuation gates $C_k \sim \text{Ber}(\alpha_\lambda^{(k)})$ and cumulative survival $B_t = \prod_{k \leq t} C_k$:

survival-chance problem for bounded $r \geq 0$ at the geometric horizon $T_\gamma \sim \text{Geom}(1 - \gamma)$

$$\max_\pi \mathbb{E}_\pi [B_{T_\gamma} r] \quad \text{subject to} \quad \Pr(B_{T_\gamma} = 1) \geq \rho$$

(SDH) survival-weighted occupancy of the transformed MDP

$$\mu_{\alpha_\lambda}^\pi(f) := \mathbb{E}_\pi \left[\sum_{t \geq 0} \gamma^t \left(\prod_{k=0}^t \alpha_\lambda(s_k, a_k) \right) f(s_t, a_t) \right] = \frac{1}{1-\gamma} \mathbb{E}_\pi [B_{T_\gamma} f(s_{T_\gamma}, a_{T_\gamma})]$$

Lagrange relaxation: single scalar η ; $\eta\rho$ is policy independent; fixed- η objective stays in SDH via $r \rightarrow r + \eta$

$$\mathcal{L}(\pi, \eta) = (1 - \gamma) \mu_{\alpha_\lambda}^\pi(r + \eta) - \eta\rho$$

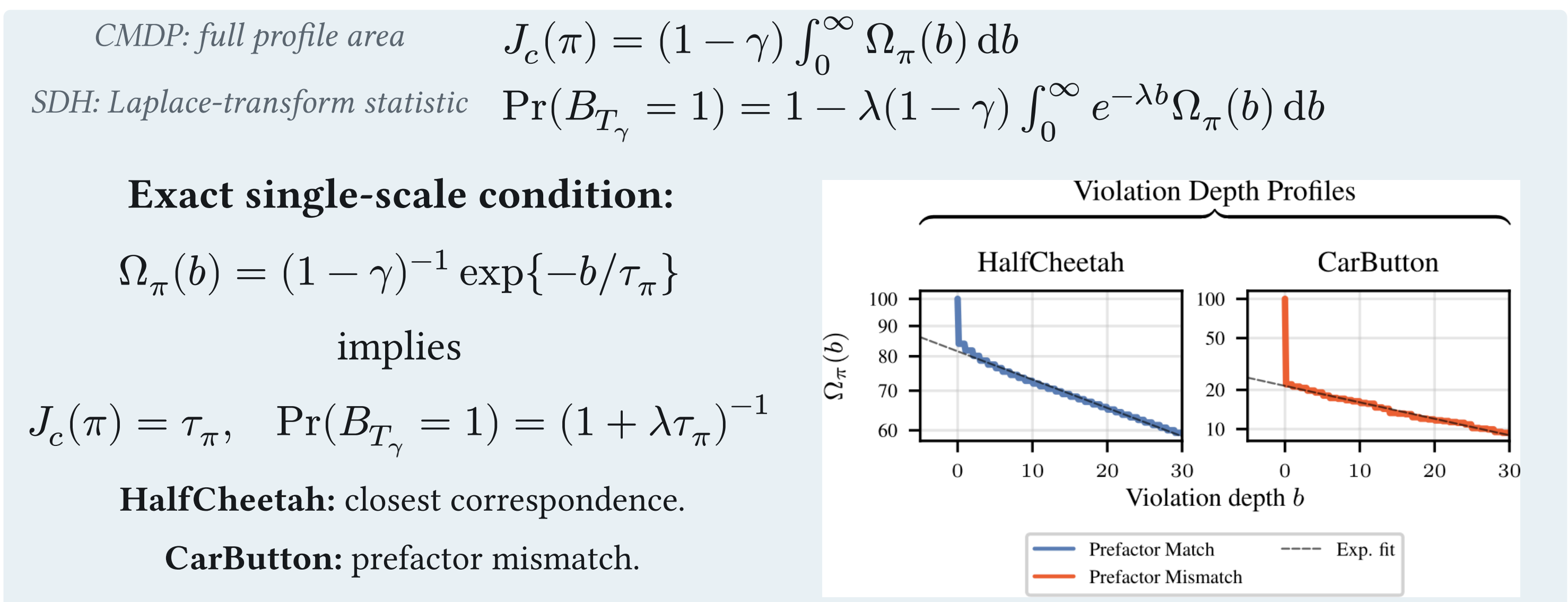
- **Violation effect:** low α_λ attenuates current reward credit and future bootstrapping.
- **Same survival gate:** one Bellman recursion; η is a survival bonus.
- **Survival:** $(1 - \gamma) \mu_{\alpha_\lambda}^\pi(1) = \Pr(B_{T_\gamma} = 1)$; so $r = 1 - f$ rewards survival even at $\eta = 0$.
- **Survival constraint:** fixed $\eta \rightarrow$ operating point; target $\rho \rightarrow$ adaptive update of dual η .

3 - Relation to CMDPs

Standard expected discounted cumulative-cost CMDP:

$$\max_\pi J_r(\pi) := \mathbb{E}_\pi \left[\sum_{t \geq 0} \gamma^t r(s_t, a_t) \right] \quad \text{subject to} \quad J_c(\pi) := \mathbb{E}_\pi \left[\sum_{t \geq 0} \gamma^t c(s_t, a_t) \right] \leq d$$

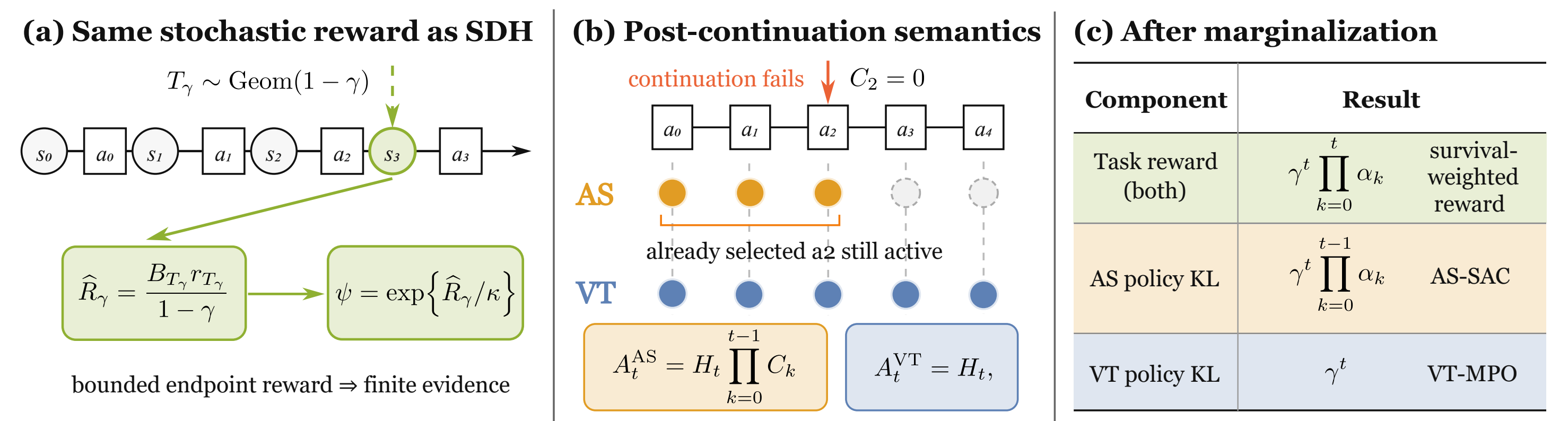
SDH is **not** a general solver for this additive-cost constraint. Cumulative violation depth $G_t := \sum_{k=0}^t c(s_k, a_k)$ and **violation-depth profile** $\Omega_\pi(b) := \mathbb{E}_\pi \left[\sum_{t \geq 0} \gamma^t \mathbb{1}\{G_t \geq b\} \right]$.



Failed single-scale correspondence limits budget-cost recovery; does not imply SDH failure.

4 - Control as Inference (CaI): post-continuation semantics

SDH tells us **which rewards survive**. But after continuation fails, **which policy decisions should still incur KL cost?**



Post-continuation semantics determine **how policy KL cost is weighted**: same **survival-weighted task objective**, different **policy KL horizons**.

$$\mathcal{J}_\sigma(\pi, \pi_0) = \mathbb{E}_\pi \left[\sum_{t \geq 0} u_t \tilde{r}_t - \kappa \sum_{t \geq 0} v_t^\sigma \ell_t \right], \quad \text{where} \quad \ell_t = \log \frac{\pi(a_t | s_t)}{\pi_0(a_t | s_t)}$$

$$\tilde{r}_t = \alpha_\lambda^{(t)}(s_t, a_t) r(s_t, a_t), \quad u_t = \prod_{k=0}^{t-1} \tilde{\gamma}_k, \quad \tilde{\gamma}_k = \gamma \alpha_\lambda^{(k)}(s_k, a_k)$$

$$v_t^{\text{AS}} = u_t \rightarrow \text{AS-SAC} \quad v_t^{\text{VT}} = \gamma^t \rightarrow \text{VT-MPO}$$

5 - Replay-compatible algorithms

- **Replay:** store raw violation signals $c = \{c_i\}$; recompute α_λ at critic update.
- **Curriculum:** λ -ramp changes α_λ and thus the SDH objective; this is not a dual update.

VT-MPO

unregularized survival task critic
 ordinary MPO actor KL cost

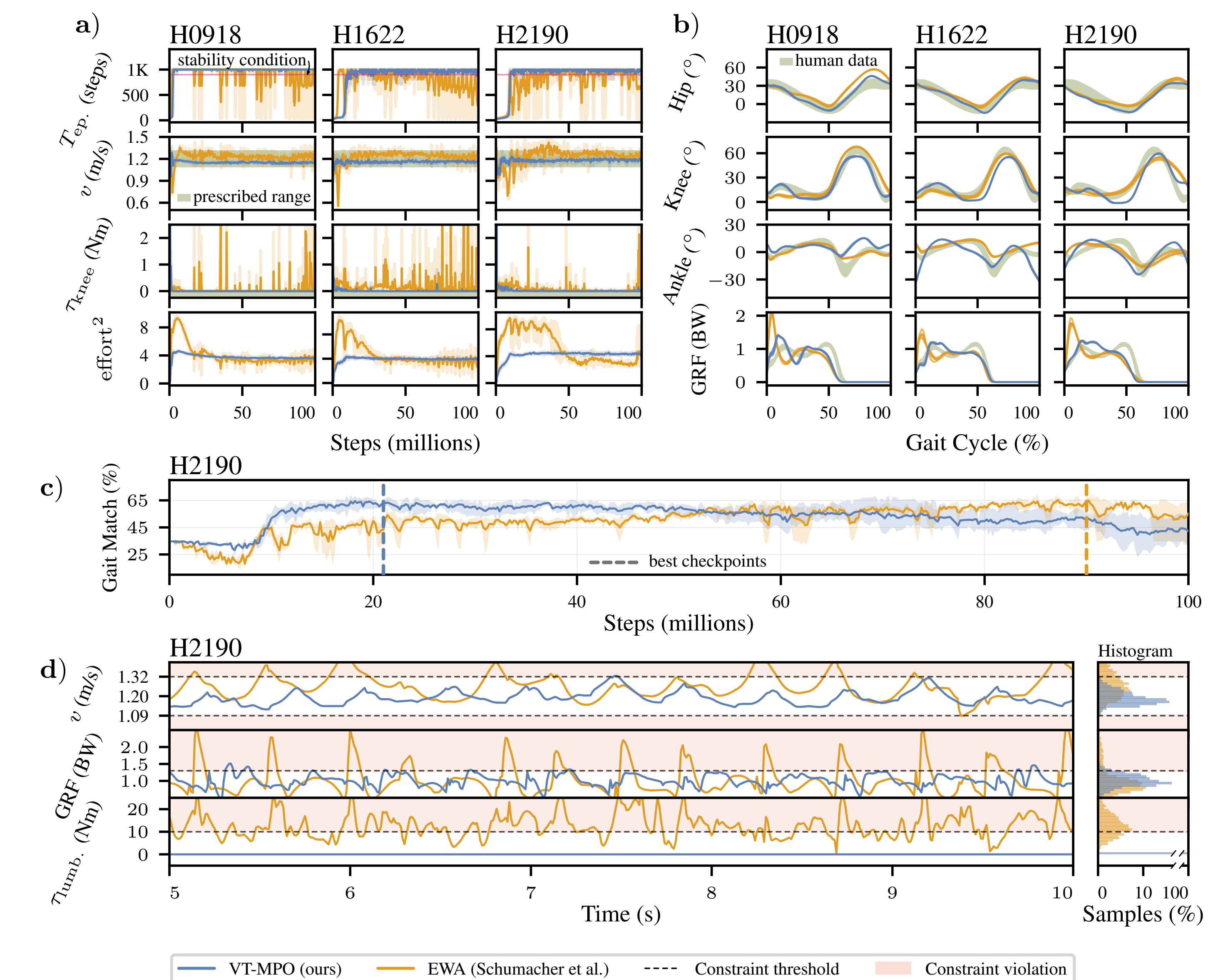
AS-SAC

KL-regularized survival critic
 survival-weighted policy KL cost

target ρ : single auxiliary survival-mass critic, regardless of the number of violation signals

6 - Hyfydy: gait optimization under local operating limits

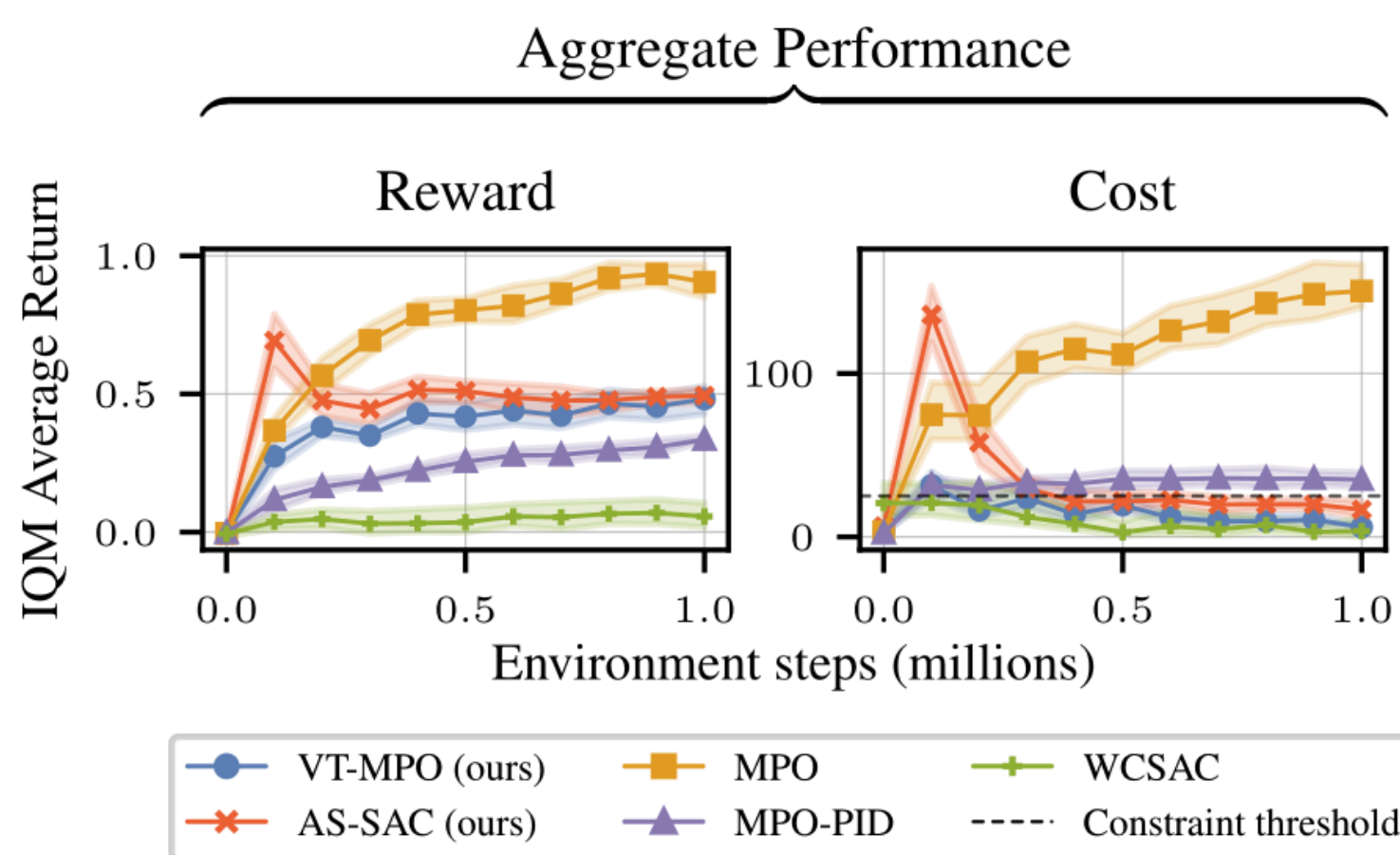
- **Domain SOTA:** EWA adapts effort penalties to balance low effort with velocity tracking.
- **SDH formulation:** formulate and solve locomotion as survival-based constrained RL with $r = 1 - \text{effort}$, maintaining target velocity under GRF, torque, and muscle-use limits.
- **Gait match:** external evaluation metric; optimized by neither method.
- **Three Scone-Hyfydy humanoids:** up to 90 muscles; fixed $\eta = 0$, no dual update.
- **Results:** all 3 walk stably by 5-10M steps; near-target speed, near-zero knee-limit violations; **smoother training**, **comparable best-observed gait match** H2190: **21M** vs. **90M**.



7 - Safety Gymnasium: a scope test

Eight budget-native tasks

- **Favorable regime:** lower violations at competitive reward on Locomotion / Circle; Ant / HalfCheetah closest to single-scale correspondence (Sec. 3).
- **CarButton:** near-exponential profile, wrong prefactor; native budget missed or reward sacrificed.
- **PointGoal:** 13% probe violations, diagnostic inconclusive. At 3M, AS-SAC is feasible without reward collapse.



Takeaway

- Constraint-dependent continuation represents a **geometric-horizon survival-chance objective** and yields **replay-compatible off-policy** AS-SAC / VT-MPO.
- In Hyfydy, we solve effort minimization under local biomechanical limits without dual optimization, yielding stable, realistic gait and markedly smoother training than EWA.
- Violation-depth profiles reveal when the SDH Laplace statistic determines cumulative cost and when that correspondence breaks.

References & resources

References

CaT: Chane-Sane et al., *Constraints as Terminations for Legged Locomotion RL*, IROS 2024.
 EWA: Schumacher et al., *Emergence of natural and robust bipedal walking by learning from biologically plausible objectives*, iScience 2025.
 CaI: Toussaint & Storkey, *Probabilistic inference for solving MDPs*, ICML 2006.
 MPO: Abdolmaleki et al., *Maximum a Posteriori Policy Optimisation*, ICLR 2018.
 SAC: Haarnoja et al., *Soft Actor-Critic*, ICML 2018.
 CaDP: Eitan Altman, *Constrained Markov Decision Processes*, Chapman & Hall/ CRC, 1999.

Resources



Website Paper Code Video